

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CM-407	Java Programming Lab	4	60	40	60

Chapter No.	Chapter/Unit Title	No. of Periods	CO's Mapped
1.	Basics, overloading, inheritance, overriding	16	CO1,CO2
2.	Streams, Interfaces and Packages and Collections.	10	CO2,CO3
3.	Exceptions and Multi threaded programming.	14	CO3,CO4
4.	Applets and Event Handling	20	CO5
Total Periods		60	

Course Objectives	i) Design object oriented programming paradigm ii) Able to develop multi tasking application with the knowledge of multi threading. iii) Familiarized to develop graphical user interface with event handling mechanism.
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Course Outcomes	CO1	Perform object oriented programming model application design.
	CO2	Design optimized definition for an application with reusability features like inheritance and polymorphism.
	CO3	Analyze modular design for real time applications by using packages concept in projects. Able to design data structures used in applications.
	CO4	Apply multi threading concepts to implement multitasking and multi programming applications.
	CO5	Develop effective dynamic user interface for any front end applications using Applets and events.

CO-PO/PSO Matrix:

CO NO.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CM-407.1	2	1	3	2		2	1	2	3	3
CM-407.2	1	3	3	3	1	3	2	2	3	3
CM-407.3	1	2	3	2	2	3	1	2	3	3
CM-407.4	1	1	3	2	2	3	2	2	3	3
CM-407.5	3	3	3	3	2	3	2	2	3	3
Average	1.5	2.6	3	2.6	1.5	3	1.6	2	3	3

3=strongly mapped, 2=moderately mapped, 1=slightly mapped

Learning Outcomes

1. Exercise programs on conditional statements and loop statements
2. Exercise programs on Strings.
3. Exercise program to create class and objects and adding methods.
4. Exercise programs using constructors and construction over loading.
5. Exercise programs on command line arguments.
 - i) Input as command line arguments and perform operation on that data.
 - ii) Input as command line arguments and update manipulated data in Files.
6. Exercise programs using concept of overloading methods.
7. Exercise programs on inheritance.
8. Write a program using the concept of method overriding.
9. Exercise on packages.
 - i) Creation of packages
 - ii) Design module to importing packages from other packages.
10. Exercise programs on interfaces.
11. Exercise programs on I/O Streams
 - i) Reading data through Keyboard
 - ii) Perform Reading and Writing operations on files using File Streams.
12. Exercise programs on Collections.
 - i) Write a java program to search a student mark percentage based on pin number using Array list.
 - ii) Write a java program to create linked list to perform delete, insert, and update data in linked list with any application.
13. Exercise on exception handling.
 - i) Programs on try, catch and finally.
 - ii) Programs on multiple catch statements
 - iii) Programs on nested try statements.
14. Exercise on multithreading
 - i) Programs on creation of single and multiple threads.
 - ii) Programs on adding priorities to multiple threads.
15. Exercise on applets
 - i) Programs on Graphics and colors.
 - ii) Simple animations using threads and graphics.
16. Exercise on AWT controls
 - i) Program to handle mouse events.
 - ii) Program to handle keyboard events.
 - iii) Programs to illustrate Text Fields and Button control.
 - iv) Programs to illustrate Check Box and List control.
 - v) Write an application program to illustrate multiple controls.

KEY COMPETENCIES

Exp. No.	Name of the experiment	Objectives	Key Competencies
1	Exercise programs on conditional statements and loop statements.	(a) Write program using if statement and switch (b) Write program using while, do and for constructs.	(a) Know the usage of IF and switch statements. (b) Compile the program and rectify the errors. (c) Observe the output.
2	Exercise programs on Strings.	(a) Write a programs to manipulate Strings (b) Write a programs to arrange array of strings in ascending order	(a) Create String objects (b) Use string class methods Observe the output.
3	Exercise program to create class and objects and adding methods.	(a) Write a program to create a class and create objects. (b) Write a program to create class adding methods and access class members.	(a) Create class. (b) Declare methods. (c) Create objects. (d) Write main method. (e) Access class members.
4	Exercise programs using constructors and construction over loading.	(a) Write a program using default constructor. (b) Write a program using parameterized constructor.	(a) Declare and define constructor. (b) Call default constructor. (c) Call parameterized constructor. (d) observe constructor overloading.
5	Exercise programs on command line arguments.	(a) Write a program to illustrate usage of command line arguments. (b) Write a program to read data as command line arguments and update it into Files.	(a) Use command line arguments. (b) Run the program. (c) Understand usage of Files. (c) Observe the output.
6	Exercise programs using concept of overloading methods.	(a) Write a program to illustrate method overloading. (b) Write a program to illustrate method overloading using constructors.	(a) Observe method overloading. (b) Overload constructor methods.
7	Exercise on inheritance.	(a) Write a program to illustrate single inheritance. (b) Write a program to illustrate multiple inheritance.	(a) Create base class. (b) Write base class constructor. (c) Create derived class. (d) Use extends keyword. (e) Use super keyword. (f) Write derived class constructor.
8	Write a program using the concept of method overriding.	Write a program using the concept of method overriding.	(a) Use method overriding. (b) Use this keyword. (c) use super keyword
9	Exercise on importing packages.	Write a program to create and importing package.	(a) Create package. (b) Use of access specifiers.

			(b) Use package. (c) Use import keyword.
10	Exercise on interfaces.	Write a program to illustrate multiple inheritance using interfaces.	(a) Define interface. (b) Use extends keyword. (c) Use implements keyword. (d) Access interface variables.
11	Exercise programs on I/O Streams	(a) Write a program to give values to variables interactively through the keyboard. (b) Write program to read and write primitive data types. (c) Write programs to handle Files.	(a) Use different data types. (b) Use read Line() method. (c) Use print In() method. (d) use File Streams Observe the output.
12	Exercise programs on Collections.	(a) Write a java program to search a student mark percentage based on pin number using Array list. (b) Write a java program to create linked list to perform delete, insert, and update data in linked list with any application.	(a) Define collection classes (b) use Array List, Linked List (c) apply List and Iterator Interface
13	Exercise on exception handling	(a) Write a program to illustrate exception handling. (b) Write a program to illustrate exception handling using multiple catch statements. (c) Write a program to illustrate exception handling using nested try.	(a) Use try – catch. (b) Use multiple catch blocks. (c) Use finally statement. (d) use Nested try
14	Exercise on multithreading	(a) Write a program to create single a thread by extending the thread class. (b) Write a program to create a single thread by implementing the runnable interface. (c) Write a program to create multiple threads. (d) Write a program to illustrate thread priorities.	(a) Use extends, new. (b) Use run() and start() methods. (c) Observe thread execution. (d) Use implements runnable interface. (e) Use set Priority() and get Priority() methods. (f) use wait(),notify() methods
15	Exercise on applets.	Write a program to create simple applet to display different shapes with colors. Write an applet program to	(a) Use <applet>...</applet> tag. (b) Add applet to html file. (c) Run the applet. (d) use graphics methods (e) use threads and graphics.

		design simple animation.	
16	Exercise on AWT controls	(a) Write an applet program to handle key events. (b) Write an applet program to handle mouse events. (c) Write an applet program to illustrate Text Field and button control. (d) Write an applet program to illustrate Check box and List control. (e) Write an applet program to illustrate multiple controls.	(a) Use keyboard event methods (b) Use mouse event methods (c) Use Text Field class methods (d) Use button class methods (e) Use Check box and List class methods